



Degree Project in Computer Science and Engineering

Second cycle, 30 credits

Adaptive Hybrid RAG

Optimizing the Cost-Performance Trade-off via Learned Query
Routing

DANTE WESSLUND

$$N = 150$$

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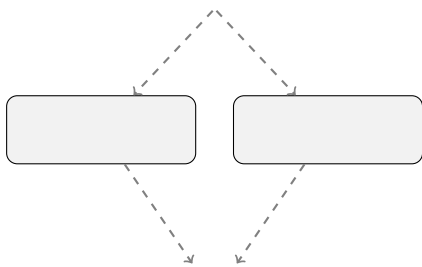
$$q\mathcal{D}_k(q)ka$$

$$p(a \mid q) = \sum_{d \in \mathcal{D}_k(q)} p_{\theta}(a \mid q, d) \, p_{\eta}(d \mid q),$$

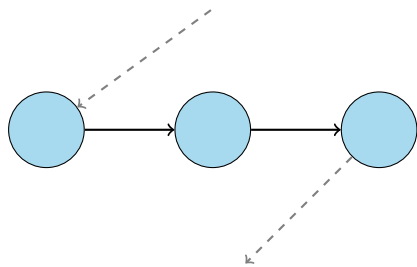
$$p_{\eta}(d \mid q)\eta p_{\theta}(a \mid q, d)\theta$$

$$X$$

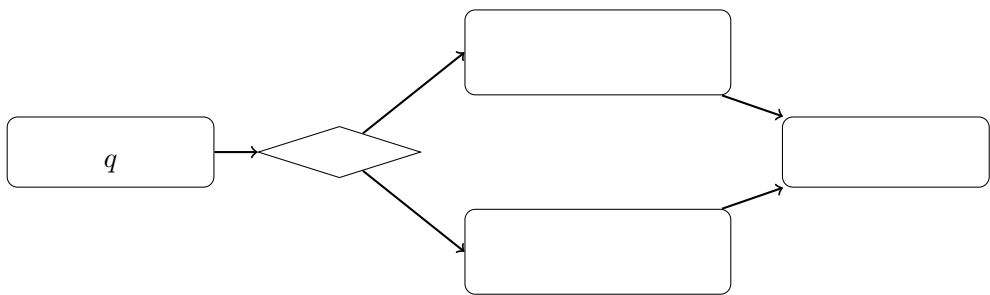
$$k$$

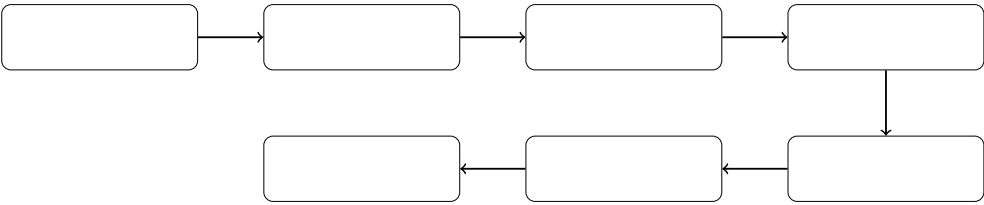


$\Rightarrow$



$\Rightarrow$





$$q_i$$

$$r_i^N=R_N(q_i),r_i^M=R_M(q_i),$$

$$\begin{array}{l} R_N R_M r_i^N r_i^M t_i^N t_i^M \\ N=150 \bar{u}^N \bar{u}^M \end{array}$$

$$\begin{array}{l} r_i^Nr_i^Nr_i^M \\ y_i\in\{0,1\} \end{array}$$

$$y_i=\left\{ \begin{array}{l} 0, \\ 1, \end{array} \right.$$

$$2\,000$$



$$X$$

$$\begin{array}{c} | \\ | \\ \hline | \\ | \\ \hline | \\ | \end{array}$$

$$q_i s_\phi(q_i) \in [0,1] \phi P_\phi(y_i=1 \mid q_i) y_i 1 q_i$$

$$V_i \in \mathbb{R}^V$$

$$s_\phi(q_i)=\sigma(\mathbb{T}_i+b), \sigma(z)=\frac{1}{1+e^{-z}},$$

$$\in \mathbb{R}^V b \in \mathbb{R} L_2$$

$$h_i \in \mathbb{R}^d d$$

$$z_i = Wh_i + b, W \in \mathbb{R}^{2 \times d}, b \in \mathbb{R}^2,$$

$$z_{i,0} z_{i,1}$$

$$s_\phi(q_i)=\frac{e^{z_{i,1}}}{e^{z_{i,0}}+e^{z_{i,1}}}.$$

$$s_\phi(q_i)=\sigma(z_{i,1}-z_{i,0})$$

$$n$$

$$\mathcal{L}(\phi) = -\frac{1}{n} \sum_{i=1}^n \alpha_{y_i} \; P_\phi(y_i \mid q_i),$$

$$\alpha_c = n/(2\,n_c) c \in \{0,1\} n_c c$$

$$s_\phi(q_i)\tau\in[0,1]\pi_\tau$$

$$\pi_\tau(q)=\begin{cases} M, s_\phi(q)\geq \tau,\\ N, s_\phi(q)<\tau, \end{cases}$$

$$MNPR$$

$$F_1=\frac{2PR}{P+R}.$$

$$r_i^Nr_i^Mt_i^Nt_i^M\pi_\tau\tau\\ m\{\cdot\}10r_M(\pi_\tau)=\frac{1}{m}\sum_{i=1}^m\{\pi_\tau(q_i)=M\}\pi_\tau C(\pi_\tau)$$

$$C(\pi_\tau)=\frac{1}{m}\sum_{i=1}^m\left[\{\pi_\tau(q_i)=N\}t_i^N+\{\pi_\tau(q_i)=M\}t_i^M\right].$$

$$U(\pi_\tau)\bar{u}^N\bar{u}^M$$

$$U(\pi_\tau)=(1-r_M(\pi_\tau))\,\bar{u}^N+r_M(\pi_\tau)\,\bar{u}^M.$$

$$(\pi_\tau)=\frac{1}{m}\sum_{i=1}^m\{y_i=1,\,\pi_\tau(q_i)=N\},$$

$$(\pi_\tau)=\frac{1}{m}\sum_{i=1}^m\{y_i=0,\,\pi_\tau(q_i)=M\}.$$

$$220,000L_2C=1.01,000$$

$$2\times 10^{-5}0.0181610\%1.0\alpha_c833\\71342$$

$$\tau^*[0.05,0.95]\tau^*\\50\%\\95\%1\,000\pm$$

$$\pi_\tau\tau$$

$$\tau_{010.01}$$

$$\begin{array}{l} 0.078 \pm 0.0190.1550.0570.076 \pm 0.009 \\ 0.6990.6670.7010.5850.8460.8030.7768,5317,13817,85752 \%60 \% \end{array}$$

$$0.5000.25495 \%1\,000\pm$$

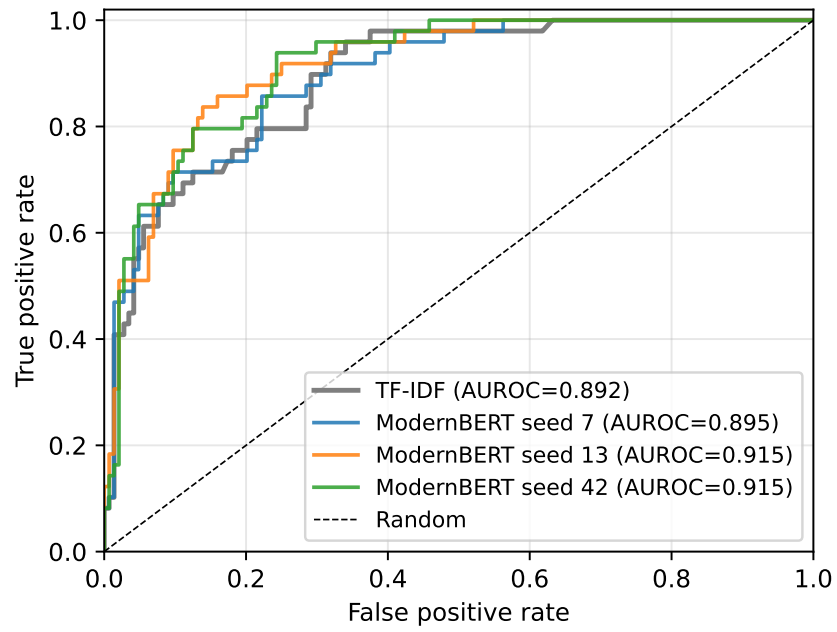
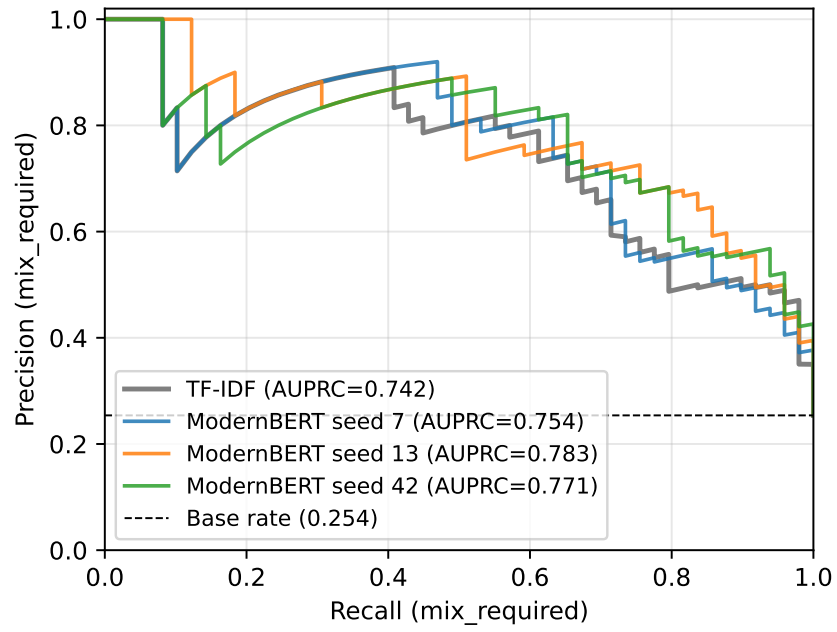
0.769 ± 0.015	0.908 ± 0.011	0.798 ± 0.012	0.701 ± 0.057	0.701 ± 0.042	0.699 ± 0.017	0.846 ± 0.016	

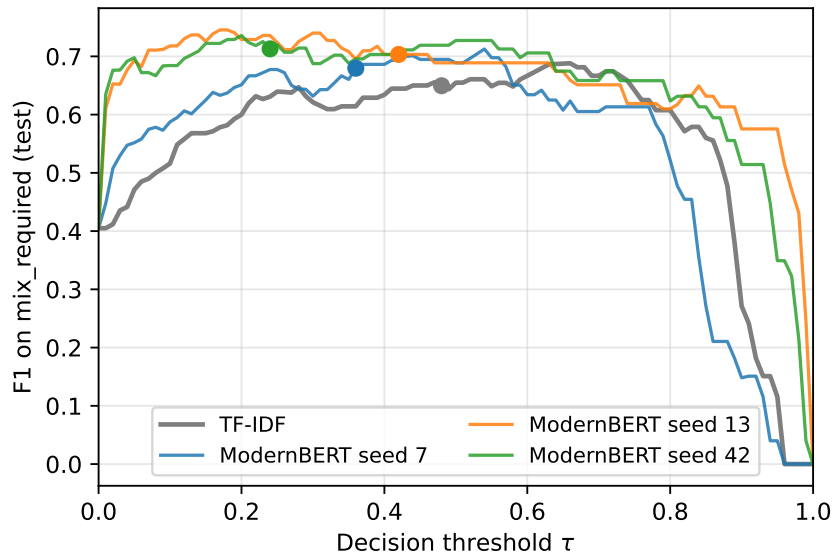
$$\bar{u}^N\bar{u}^MN=150(17,857-)/17,8570.254\pm$$

0.256 ± 0.027	14.92 ± 0.69		0.076 ± 0.009	0.078 ± 0.019		

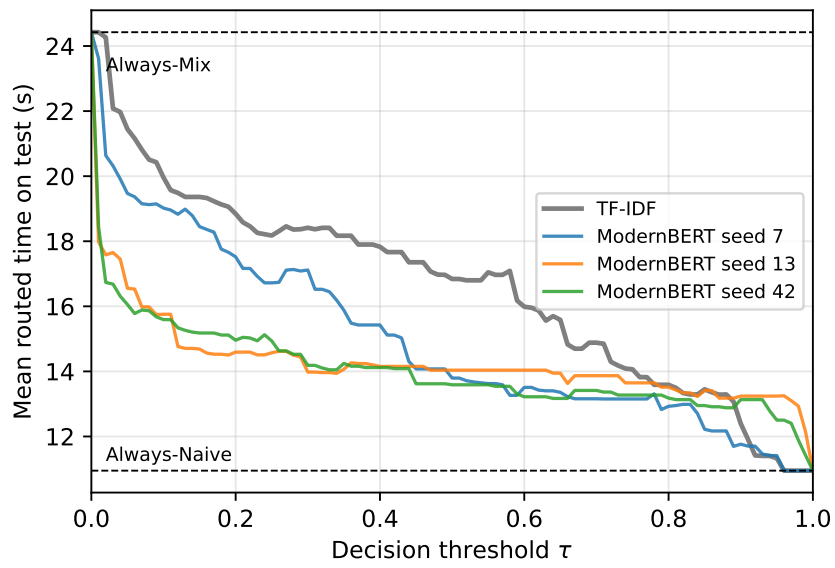
$$\begin{array}{l} \tau^{\star}=0.24,0.36,0.42427130.6990.017 \\ 01\tau=0\tau=1 \\ \tau \\ 2908333865150.8390.8140.7990.7530.3750.1670.4180.3630.3160.2110.345 \\ 0.2730.7070.606 \end{array}$$

$$\begin{array}{l} 0.8390.8140.7530.7990.4180.316 \\ 0.667 \end{array}$$

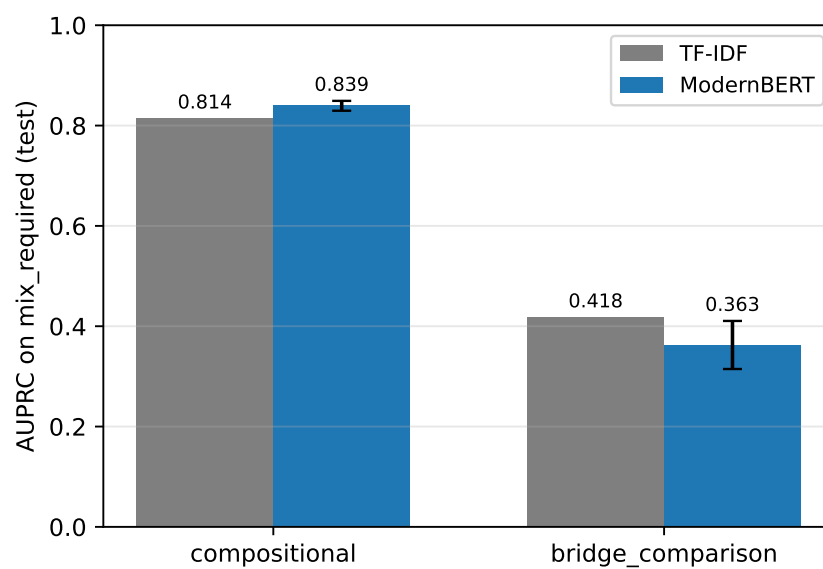
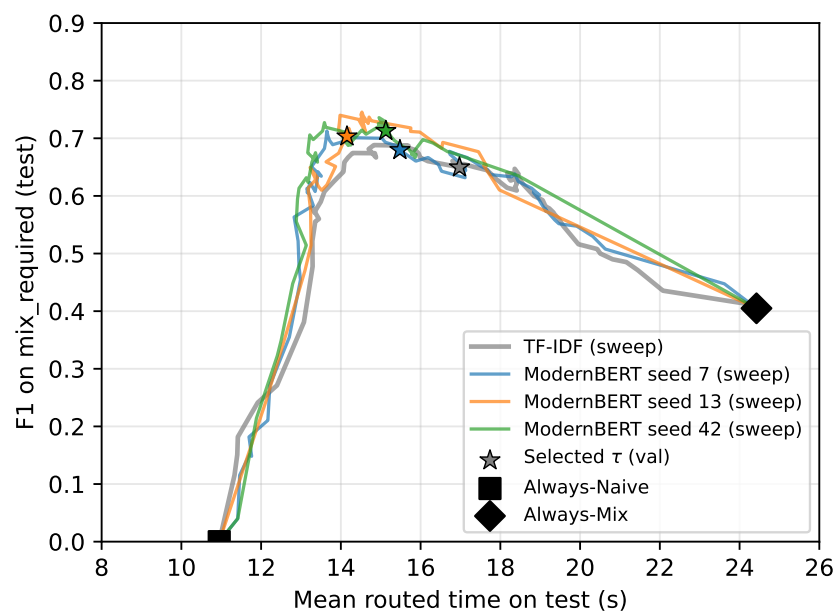


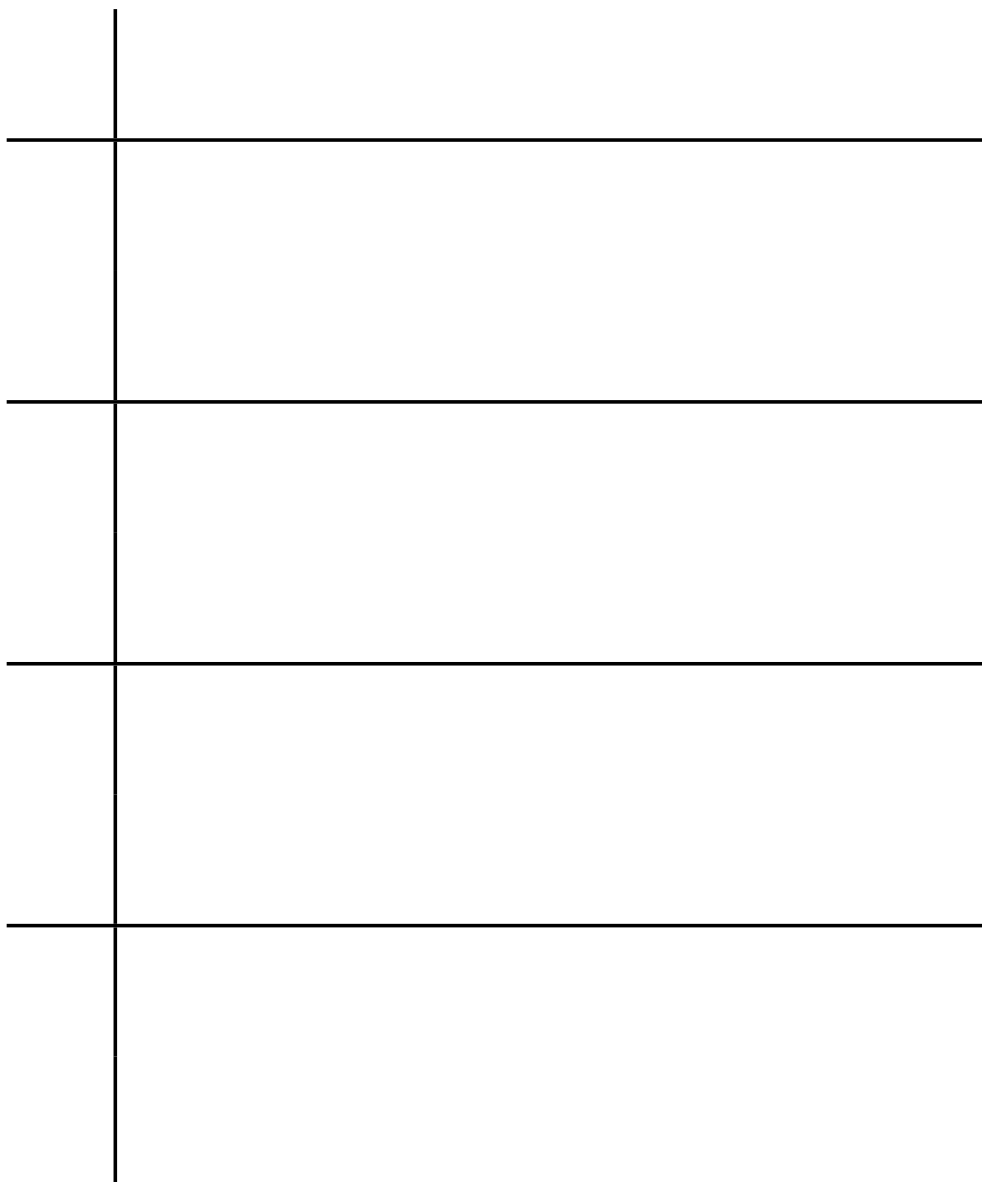


$$\tau^* = 0.4842 \tau^* = 0.247 \tau^* = 0.3613 \tau^* = 0.42$$



193[0.836, 0.936]0.908193 ≤ 0.0170.0570.042193
 10.9524.423,45817,85715173039 %7,1008,5005260 %





1930.7690.7420.7980.7840.6990.6500.8460.788≤ 0.0170.0570.076 ±
0.0090.078 ± 0.0190.155
151710.9524.427,1008,5003,45817,85760 %52 %
0.6670.7940.803

$N = 1501,286$
1,286

